

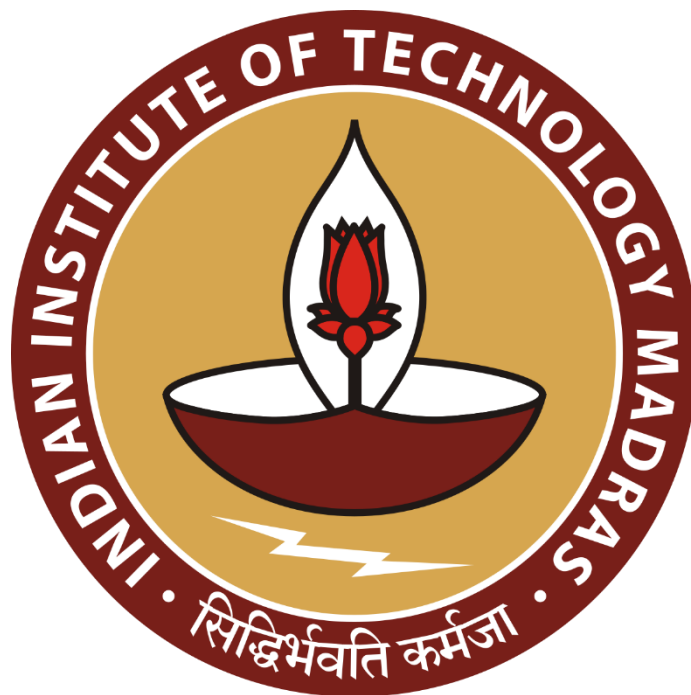
Optimizing Inventory and Expanding Product Range at Vishnu Priya Variety Store

A Proposal report for the BDM capstone Project

Submitted by

Name: Sreyash Sourav

Roll number: 24F2001076



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project titled “**Optimizing Inventory and Expanding Product Range at Vishnu Priya Variety Store**”. I extend my appreciation to **Mr. Pratap Chandra Behera**, for providing necessary resources and enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analysed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

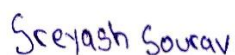
I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me.

In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate



Name: Sreyash Sourav

Date: 01 Oct, 2025

1 Executive Summary

‘Vishnu Priya Variety Store’ is a retail grocery store located in Bhubaneswar, Odisha. Established in 2025, it is general grocery store that sells goods like Rice, Pulses, Dairy Products, Edible Oil and also some other essential commodities. It operates under (B2C) business model, directly serving customers products for their everyday needs.

The main problems faced by the shop are inventory management and limited product variety. Some items remain unsold for a long time or are purchased in extra quantity. This causes wastage, loss of money and storage issues. The second problem is limited range of products. The shop currently offers only a few types of grocery items, which limits customer choice and also affects sales. As a result, fewer people visit the store and the growth remains slow.

This project aims to help a grocery shop manage inventory and sales using data. It focuses on reducing deadstock and overstocking, and to introduce new variety of products to improve sales. I will use tools like Excel and Google sheets for storing and organising sales and data. Simple charts and graphs help visualize trends, enabling the store to make better decision and improve the sales.

2 Organization Background

Shop Name: Vishnu Priya Variety Store

Address: K1-388, Ghatikia, Kalinga Nagar, Bhubaneswar – 751003

Owner's Name: Mr. Pratap Chandra Behera

‘Vishnu Priya Variety Store’ is a newly established grocery store located in Bhubaneswar, Odisha. The store was started this year by Mr. Pratap who previously worked in a textile industry. His father Mr. Rudra Behera had a successful grocery business in their native village which gave him valuable insights and experience about managing a grocery store. The store primarily sells everyday grocery items such as rice, pulses, edible oil, dairy products and other essential commodities. Although it is new, ‘Vishnu Priya Variety Store’ aims to provide quality products and convenient shopping

to the local community. Following a B2C model, the store directly serves individual customers and households in the surrounding area.

3 Problem Statement

3.1 The store faces inventory management problems such as overstocking and deadstock, where some products remain unsold for long periods leading to wastage. Poor tracking of stock levels causes financial losses, storage issues, and difficulty in understanding which products are in demand, as result it reduces overall efficiency and affect profitability.

3.2 The shop currently sells only a limited range of grocery items, which restricts customer choices and reduces sales. Due to this, fewer people visit the store and overall sales remain low, which affects the store's ability to compete with other local shops.

4 Background of the Problem

'Vishnu Priya Variety Store', owned by Mr. Pratap Chandra Behera, is a newly established grocery shop in Bhubaneswar, Odisha. As a new business, the store is facing challenges in managing operations effectively. One of the major problems is inventory management. Mr. Pratap is currently not maintaining proper records or using a **khata** or digital notes, which makes it difficult to track which products are selling and which are not. This leads to issues such as overstocking and deadstock, where some items remain unsold for long periods. Without proper tracking it is really difficult to understand product demand, plan purchases or make business decisions.

Another significant problem is the limited variety of products. The store primarily sells basic grocery items, which restricts customer choices and reduces sales. Since a school (**DAV, Kalinga Nagar**) is located nearby, there is a high demand for stationery products like books, notebooks, pens and other related items.

These problems directly affect the store's growth, customer satisfaction and profitability. Addressing these issues is crucial for operational efficiency, reducing losses and creating opportunities for business expansion. Proper record keeping and data analysis can provide Mr. Pratap with insights to manage inventory better and understand customer needs, founding the formation for long term success.

5 Problem Solving Approach

To address the challenges faced by 'Vishnu Priya Variety Store', a data driven approach will be used to solve two major problems – Inefficient Inventory Management and Limited Product Variety. Rather than relying on assumptions and intuition I will use business data to make informed decisions.

1. Data collection and Monitoring:

Analysis: Firstly, I will collect accurate data on stock levels, sales and customer demand. I will use tools like Excel and Sheets to record daily sales and inventory details. By maintaining a digital record instead of manual notes, Mr. Pratap can easily track which products sell fast and which remain unsold for long periods. This will help identify deadstock and overstocking patterns. Monitoring data over time will give clear insights into demands and customer behaviour, which will support better stock planning. This will also help him understand seasonal demand for specific products.

2. Data Analysis for Inventory Management:

Statistical Calculations: Once I collect the data, I will be using charts, tables and basic statistics to analyse the trends. The analysis will help determine reorder levels and avoid unnecessary bulk purchases.

- **Trend Analysis:** I will study sales data over time to identify which products sell fast and which remains unsold, helping in better demand forecasting.
- **Reorder Level Setting:** I will use data to decide when and how much to restock each product, ensuring continuous availability without overstocking.

- **Waste Reduction:** Detect slow moving or expiring products early to reduce wastage and improve space utilization.

Through this systematic data analysis ‘Vishnu Priya Variety Store’ can achieve smoother inventory control, better resource utilization and improved profitability.

3. Integrating Owner Insights with Data for Business Optimization:

After analysing the sales and inventory data, further steps involve identifying business gaps, evaluating suppliers and ensuring continuous improvement. To understand the store’s challenges better I had a conversation with Mr. Pratap, who shared that he often faces issues with product shortages, delayed deliveries and lack of supplier coordination.

- **Identifying Gaps:** After analysing the data, I observed which products are frequently demanded by the customer but are missing from the store, helping identify opportunities for product expansion.
- **Supplier Evaluation:** By comparing supplier data such as cost, delivery, time and product quality, the store can select the most dependable and cost-effective suppliers.

This collaborative and data driven approach will lead to smarter decisions and sustainable growth for ‘Vishnu Priya Variety Store’.

6 Expected Timeline

6.1 Work Breakdown Structure:

- Discussion with the shop owner: 29 Jun – 30 Jun
- Data Collection: I collected data for nearly 60 days (July and August)
- Data Cleaning and Processing: I have successfully cleaned the data by mid of September.
- Proposal Preparation: I prepared my project proposal by the starting of October.
- Data Analysis: I started basic data analysis by the mid of September.
- Complete the analysis on how to solve the problem and upload my final project: 27 Nov

6.2 Gantt Chart:

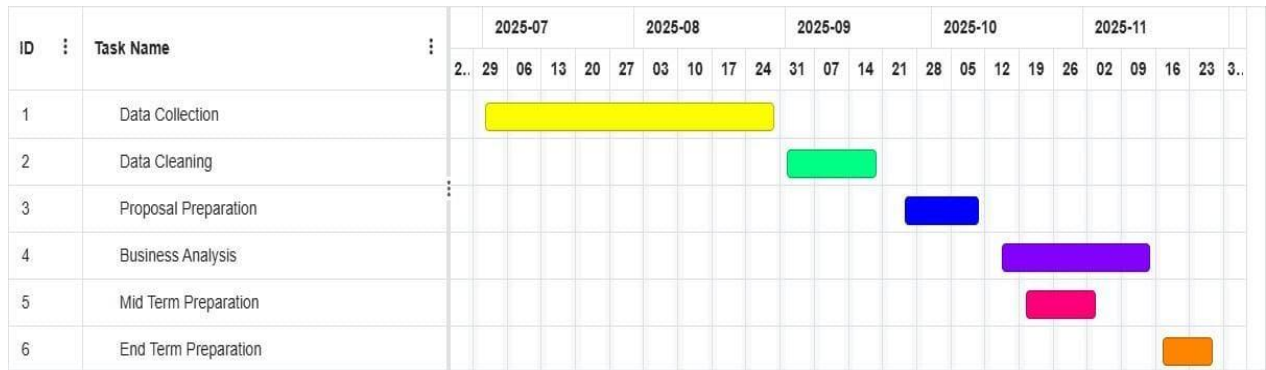


Figure 1 Expected timeline for completion of project

7 Expected Outcome

- The analysis will minimize overstocking and deadstock by accurately tracking inventory and analysing sales patterns.
- It will identify gaps in product offerings and recognise high demanded items to expand the product range.
- It will help the owner make informed, data driven decisions for purchasing, stocking and introducing new products.
- Improve customer satisfaction through consistent product availability and a wider variety of choices.